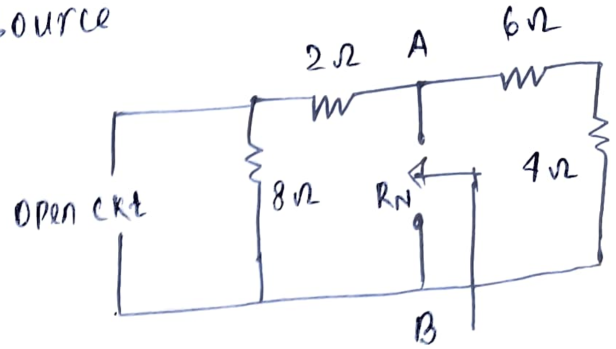
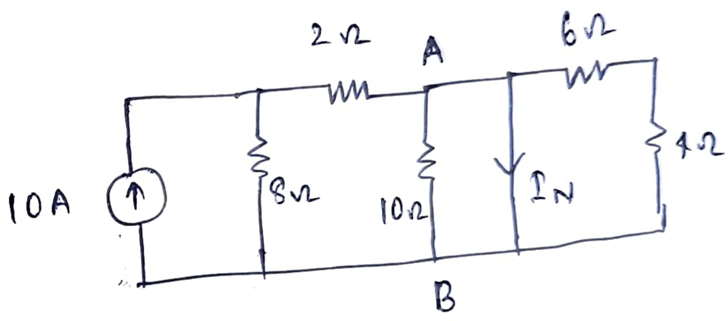
1. <u>Step 1.</u> Find Norton Resistance (R_N) :→ We need to remove the load resistance across ~~load~~ A-B terminal and deactivate the current source



$$R_N = (10\Omega) \,||\, (10\Omega)$$

$$R_N = \frac{10 \times 10}{10 + 10} = 5\Omega$$

<u>Step 2</u> Find Norton current (I_N) :→ short circuit the A-B terminal



So, no current will flow across 10Ω resistor, 6Ω resistor & 4Ω resistor

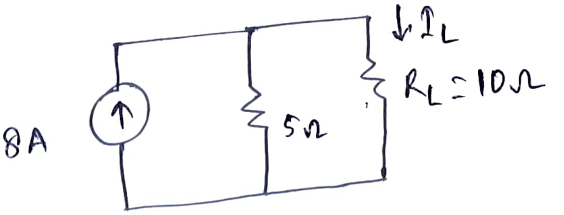
So, I_N will be same as current flowing 2Ω resistor

Using current divider rule, current through 2Ω resistor is $\frac{10 \times 8}{8 + 2} = 8\,A$

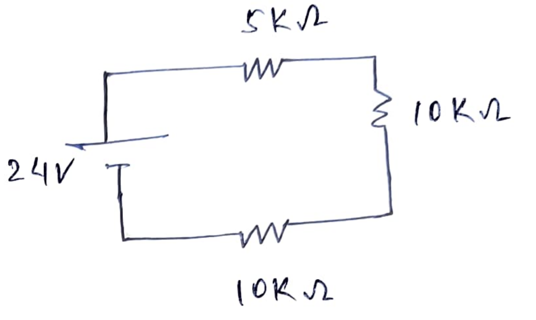
<u>Step 3</u> Draw the equivalent Norton circuit and find out current value across load

Use current Divider Rule

$$I_L = \frac{8 \times 5}{5+10} = \frac{8}{3} \text{ A}$$



Q. 2. Using voltage Divider Rule



voltage across $10 k\Omega$ is

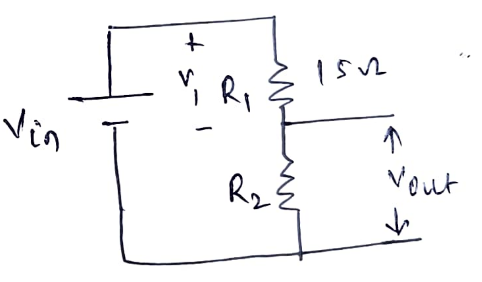
$$\frac{24 \times 10}{5+10+10} = 9.6V$$

Q. 3 Given $V_{in} = 50V$

$V_{out} = 20V$

$R_1 = 15\Omega$

$R_2 = ?$



so, $V_{in} = V_1 + V_{out}$

$50 = V_1 + 20$

$V_1 = 30V$

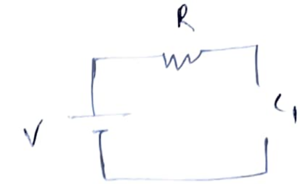
current through $R_1 = \frac{V_1}{R_1} = \frac{30}{15} = 2A$

R_1 and R_2 are connected in series, so, current through $R_2 = 2A$

so, $R_2 = \frac{V_{out}}{2} = \frac{20}{2} = 10\Omega$

Q. 4. (a)

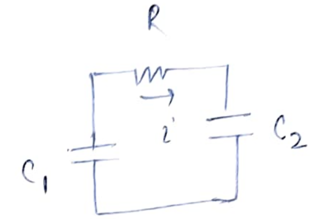
Under steady state



current in $C_1 = 0$

current in $R_1 = 0$

voltage across $C_1 = V$



(b) $V_{C_1} = Ri + V_{C_2}$

$\Rightarrow V_{C_1} - Ri - V_{C_2} = 0$ — (A)

so, $i = C_2 \dfrac{dV_{C_2}}{dt}$ — (B)

or $i = -C_1 \dfrac{dV_{C_1}}{dt}$ — (C)

so, Differentiating eqn (A)

$$\frac{dV_{C_1}}{dt} - R\frac{di}{dt} - \frac{dV_{C_2}}{dt} = 0$$

$$\frac{-i}{C_1} - R\frac{di}{dt} - \frac{i}{C_2} = 0$$

so, $$\boxed{i\left(\frac{1}{C_1} + \frac{1}{C_2}\right) + R\frac{di}{dt} = 0}$$

(c) If $C_1 = C_2 = C$

then $$R\frac{di}{dt} + \frac{2i}{C} = 0$$

$$R\frac{di}{dt} = -\frac{2i}{C}$$

so, $$\int \frac{1}{i}\,di = \int -\frac{2}{RC}\,dt$$

$$\ln i = -2t/RC + \ln K$$

$$\ln\left[\frac{i}{K}\right] = \frac{-2t}{RC}$$

$$\frac{i}{K} = e^{-2t/RC}$$

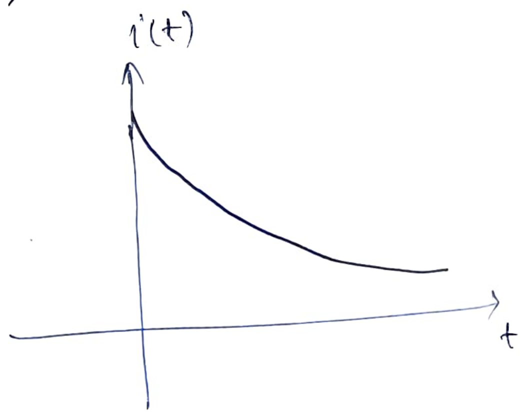
$$i = Ke^{-2t/RC} \quad \text{—— (1)}$$

Also, $i = \dfrac{V_{c_1} - V_{c_2}}{R}$

at $t = 0^+$, $V_{c_2} = 0$ (capacitor opposes sudden change in voltage)

so, $i = \dfrac{V_{c_1}}{R} = \dfrac{V}{R}$

so, (1) $\Rightarrow$ $\boxed{i = \dfrac{V}{R} e^{-2t/RC}}$ $t > 0$

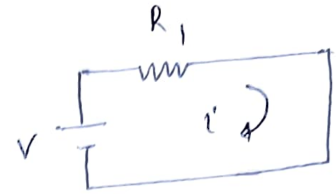


(5)

Step 1

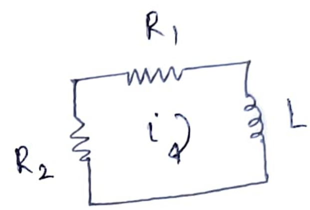
Draw the circuit when the switch was at position 1. (inductor will become short circuit in steady state on DC supply)

$$i_0 = \frac{V}{R_1}$$



Step 2

Draw the circuit when switch is at position 2



Step 3

find τ

$$\tau = \frac{L}{R_1+R_2}$$

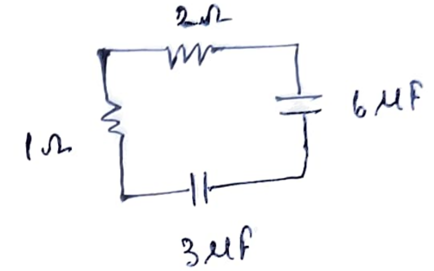
Step 4

Write down current expression

$$i = i_0 e^{-t/\tau}$$

$$\boxed{i = \frac{V}{R_1} e^{-(R_1+R_2)t/L}}$$

6. (a) Draw the equivalent circuit

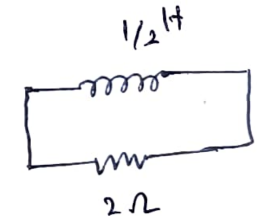


$R_{eq} = R_1 + R_2 = (1+2) = 3\,\Omega$

$C_{eq} = \dfrac{C_1 C_2}{C_1 + C_2} = 2\,\mu F$

$\tau = RC = (3 \times 2)\,\mu s = 6\,\mu s$

(b) Draw the equivalent circuit



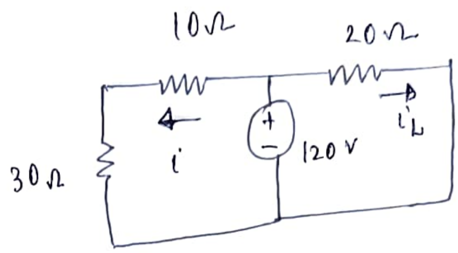
$\tau = L/R = \dfrac{1}{2 \times 2} = 0.25$ sec

(7) Step 1

Find the initial current through inductor when switch was closed (inductors will become short circuit in DC steady state)

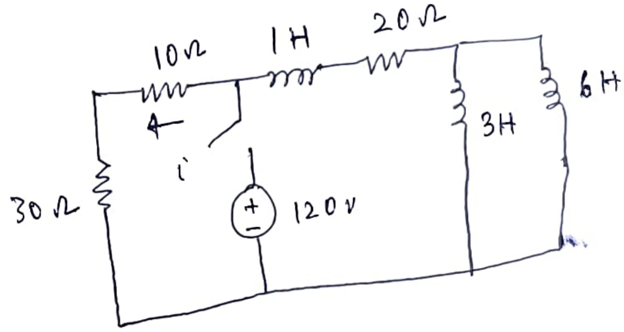
$i = \frac{120}{40} = 3\,A$

$i_L = \frac{120}{20} = 6\,A$



Step 2

Draw the circuit once switch is open



Step 3

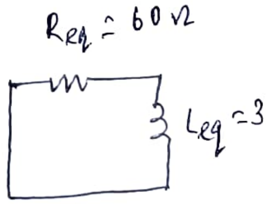
Find out τ

$\tau = \frac{L_{eq}}{R_{eq}}$

$L_{eq} = 1 + (3 \| 6) = 3\,H$

$R_{eq} = (20 + 10 + 30)\,\Omega = 60\,\Omega$

$\tau = \frac{3}{60} = \frac{1}{20}$ sec

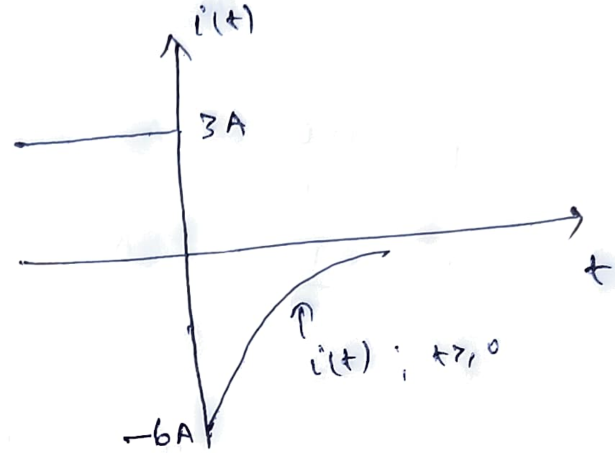


Step 4

Write down current expression

$I = I_0 e^{-t/\tau} = 6e^{-20t}$

But given i is in opposite direction of calculated initial current so $\boxed{i = -I = -6e^{-20t} \ \text{A}}$ $t \geq 0$



8. Ans. (a) We know that current through inductor can be written as

$$I = I_0 e^{-t/\tau} \quad \text{——— (1)}$$

$R = 10\,\Omega$ (Given)

$P_1 = I_1^2 R$

$P_2 = I_2^2 R$

$$\frac{P_1}{P_2} = \frac{I_1^2}{I_2^2}$$

$P_2 = 0.5\,P_1 \longrightarrow$ eq as per question

so, $I_2 = \sqrt{0.5}\, I_1$

so, (i) $\Rightarrow \sqrt{0.5}\,\cancel{I_1} = \cancel{I_1}\, e^{-t/\tau}$

$\ln(\sqrt{0.5}) = -t/\tau$

$t/\tau = 0.347$

put $t = 1$ sec (According to question)

so, $\frac{1}{\tau} = 0.347 \Rightarrow \tau = 2.88$ s

$$\tau = \frac{1}{0.347} \qquad \Rightarrow L = \frac{R}{0.347} = \frac{10}{0.347} = 28.82\,H$$

$$\frac{L}{R} = \frac{1}{0.347}$$

8. (b)

so, $\because$ 10 H introduces $0.5\,\Omega$ resistance

$\therefore$ 1 H " $\frac{0.5}{10}\,\Omega$ "

$\therefore$ L " $\frac{0.5L}{10}\,\Omega$ "

so, $R_{eq} = R_L + 10 = \left(\frac{0.5L}{10} + 10\right)\Omega$

so, we know that

$$\tau = \frac{L}{R_{eq}}$$

$$\tau = \frac{L}{\left(\frac{0.5L}{10} + 10\right)}$$

$$\Rightarrow \tau = \frac{L \times 10}{0.5L + 100}$$

$$\Rightarrow \tau(0.5L + 100) = 10L$$

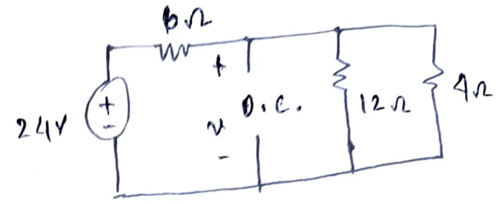
put $\tau = 2.88$ sec from (a)

$$\boxed{L = 33.64\ H}$$

(9) since both wheels are same in appearance and dimension, they have similar mass and inertia. Still one wheel has a larger time constant (twice), this is because it has lesser friction compared to wheel. Therefore, it will be easier to pedal. so, the wheel with larger time constant will be easier to padel.

(10)

<u>Step 1</u> Form the ckt before switch was open



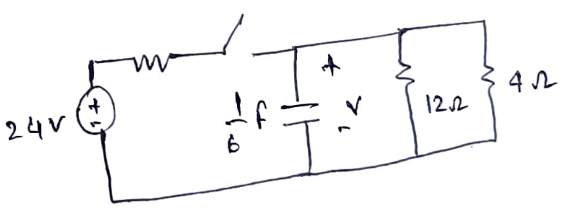
The capacitor will be open circuit in steady state for DC supply

Voltage across capacitor $V_c = \frac{24 \times (12 \| 4)}{(12 \| 4) + 6}$

as 12 Ω and 4 Ω are connected in parallel and their equivalent will be 3 Ω

so, $V_c = \frac{24 \times 3}{3+6} = 8V = V_0$.

<u>Step 2</u> Draw the ckt after switch is open



<u>Step 3</u> Find out τ

$\tau = R_{eq} \times C$

R_{eq} across c will be $(12 \| 4)\,\Omega = 3\,\Omega$

so $\tau = 3 \times \frac{1}{6} = 0.5$ sec

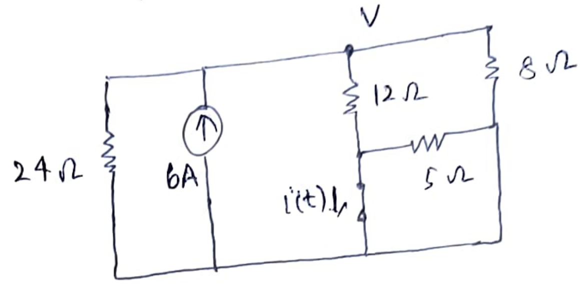
<u>Step 4</u> Write the voltage expression across capacitor

$V_c(t) = V_0 e^{-t/\tau}$

$$\boxed{V_c(t) = 8e^{-2t}}$$ $t \geq 0$

(11) Step 1 — find the initial current through inductor when switch was closed (inductor will be short circuit in steady state for DC supply)

Apply nodal analysis at V



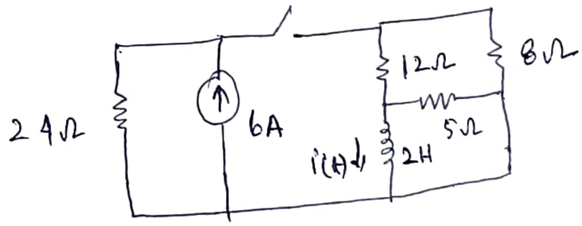
$$-6+\frac{V}{24}+\frac{V}{12}+\frac{V}{8}=0 \quad \text{—— (1)}$$

There will be no current through 5 Ω resistor as it is short circuit

so, solve eqn (1) ⇒ $V = 24\,V$

so, current through inductor will be $\frac{V}{12} = 2\,A$

Step 2 — Draw the circuit once switch is open



Step 3 — find out τ

$\tau = L/R$

$R_{eq} = (12+8) \,||\, 5 = 4\,\Omega$

so, $\tau = \frac{2}{4} = 0.5$ sec

Step 4 Write down current expression through inductor

$i(t) = i_0 e^{-t/\tau}$

$\boxed{i(t) = 2e^{-2t}}$ $t \geq 0$